

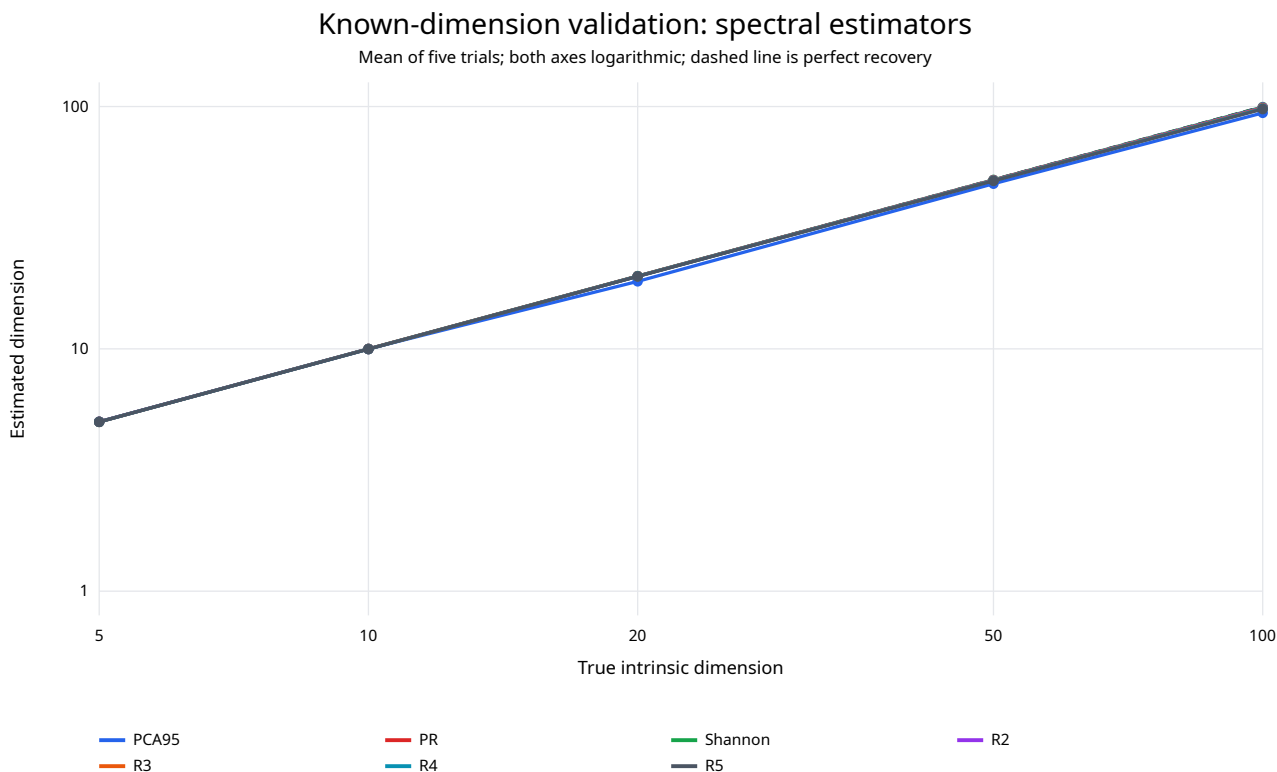
EffDim validation on known-dimensional manifolds

Five independent isotropic Gaussian datasets were generated at each true dimension (5, 10, 20, 50, and 100), with 10,000 samples embedded into a random 256-dimensional linear subspace. Exact GPU neighbours used $k=10$.

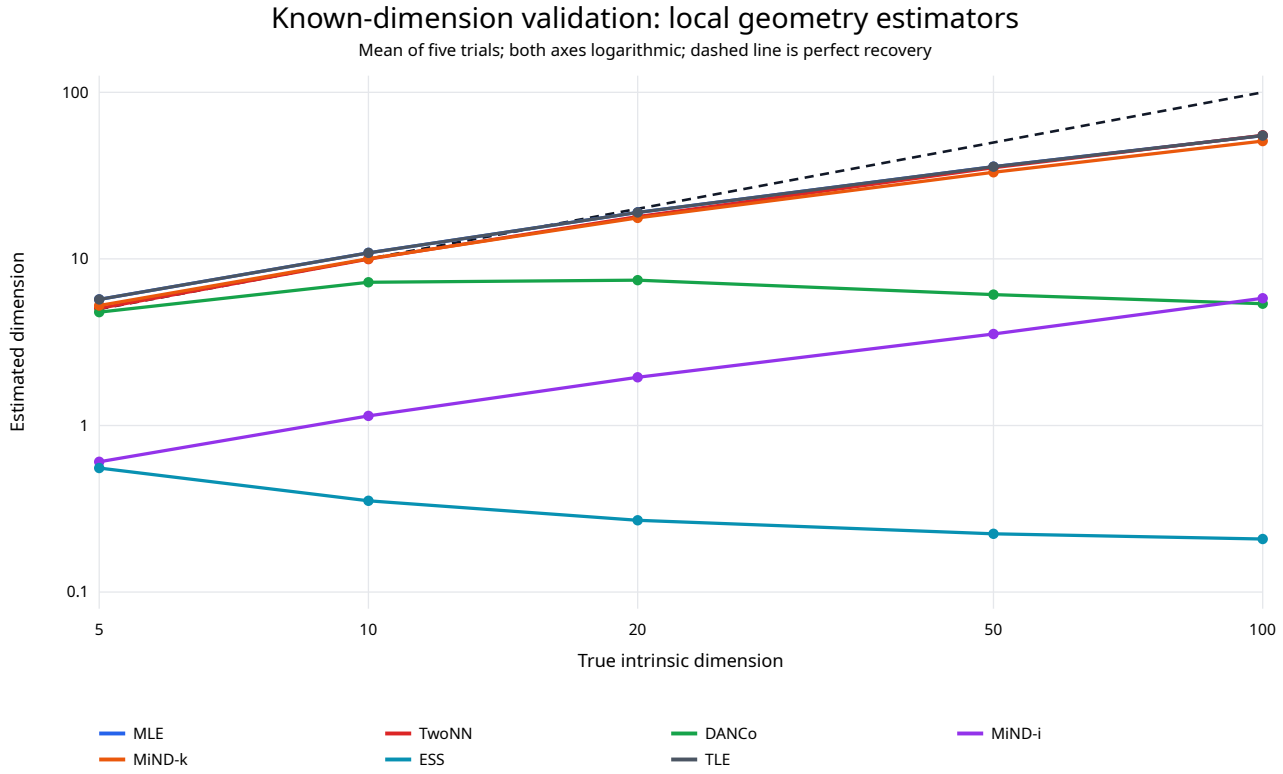
Main findings

1. **Participation ratio, Shannon effective rank, and Rényi dimensions recover the known linear rank almost exactly.** Even at dimension 100 their means lie between 97.6 and 99.5.
2. **PCA-95 behaves according to its definition:** 5, 10, 19, 48, and 94.2. It is a 95%-variance rank rather than an unbiased intrinsic-dimension estimate.
3. **MLE, TLE, Two-NN, and MiND-MLk work through dimension 20 but underestimate high dimensions severely.** At true dimension 100 they return about 51–55, showing finite-sample and $k=10$ limitations.
4. **DANCo saturates near 5–7, MiND-MLi remains far too low, and the simplified ESS statistic decreases toward 0.2.** These implementations should not be selected as general dimension estimators from stability alone.
5. **Geometric-mean ED is numerically invalid on rank-deficient embeddings.** Null covariance directions drive enormous values instead of the known rank.

Spectral estimators



Local geometry estimators



Mean estimates

Method	d=5	d=10	d=20	d=50	d=100
PCA-95	5.000	10.000	19.000	48.000	94.200
Participation ratio	4.998	9.990	19.960	49.736	99.007
Shannon ED	4.999	9.995	19.980	49.867	99.499
Rényi $\alpha=2$	4.998	9.990	19.960	49.736	99.007
Rényi $\alpha=3$	4.997	9.986	19.941	49.606	98.525
Rényi $\alpha=4$	4.995	9.981	19.921	49.477	98.056
Rényi $\alpha=5$	4.994	9.976	19.902	49.351	97.600
Geometric mean ED	6.472e+14	3.248e+14	8.056e+13	1.221e+12	1.288e+09
MLE	5.716	10.862	19.012	35.808	55.022
Two-NN	5.045	9.967	17.932	35.263	55.199
DANCo	4.793	7.242	7.449	6.100	5.386
MiND-MLi	0.605	1.142	1.945	3.538	5.803
MiND-MLk	5.248	10.023	17.613	33.147	51.017
ESS	0.555	0.353	0.270	0.224	0.208
TLE	5.716	10.862	19.012	35.808	55.022